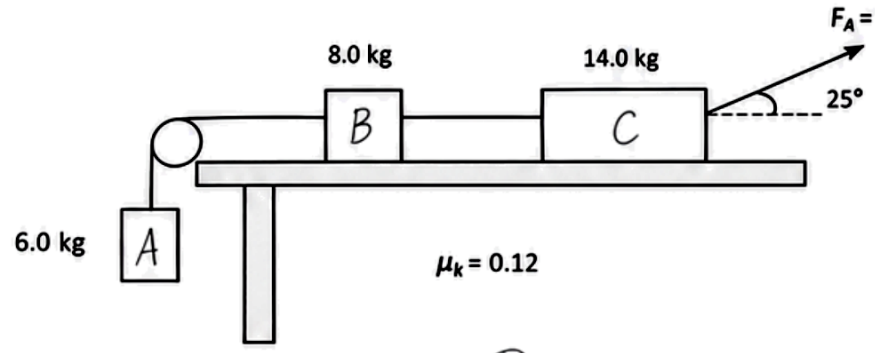
2.5 Complex Systems

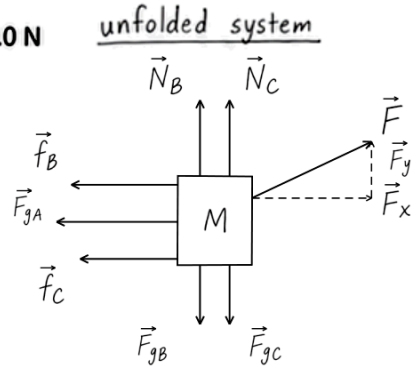
Learning Goals (Students will be able too…)

- Solve complex problems involving inclined planes and pulleys combined.

Putting It All Together

1. A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the 8.0 kg and 14.0 kg blocks with the table is 0.12. Calculate:
   a) the acceleration of the 8.0 kg block [0.39 m/s^2]
   b) the tension in the string between the 8.0 kg and 14.0 kg block. [74 N]





+x: A up; B and C right.

① y, block B:

$0 = N_B - m_B g$

$N_B = m_B g$

$N_B = (8.0)(9.81)$

$N_B = 78.48\ N$

② y, block C:

$0 = N_C + F\sin\theta - m_C g$

$N_C = m_C g - F\sin\theta$

$N_C = (14.0)(9.81) - (100.0)\sin 25°$

$N_C = 95.08\ N$

③ x, whole system:

$M = m_A + m_B + m_C$

$F_{net,x} = F\cos\theta - f_B - f_C - m_A g = Ma$

$$a = \frac{F\cos\theta - \mu_k(m_B g + m_C g - F\sin\theta) - m_A g}{m_A + m_B + m_C}$$

$$a = \frac{100.0\cos 25° - 0.12\,(8.0(9.81) + 14.0(9.81) - 100.0\sin 25°) - 6.0(9.81)}{6.0 + 8.0 + 14.0}$$

$a = 0.39085\ m/s^2$

$a = +0.39\ m/s^2$

$\vec{a}_B = 0.39\ m/s^2$ [positive x]

+x reversed: A down, B/C left.

$$a_2 = \frac{m_A g - F\cos\theta - \mu_k(N_B + N_C)}{M}$$

$$= \frac{6.0(9.81) - 100.0\cos 25° - 0.12\,(78.48 + 95.08)}{6.0 + 8.0 + 14.0}$$

$= -1.88\ m/s^2$ Opposite to this case's +x.

b) x, block C:

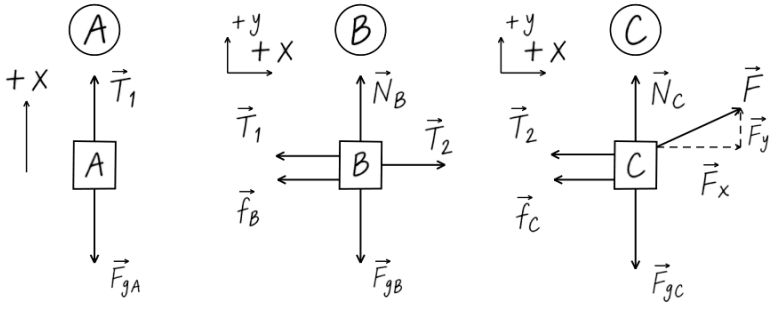
$m_C a = F\cos\theta - \mu_k N_C - T_2$

$T_2 = F\cos\theta - \mu_k(m_C g - F\sin\theta) - m_C a$

$T_2 = 100.0\cos 25° - 0.12(14.0(9.81) - 100.0\sin 25°) - 14.0(0.39085)$

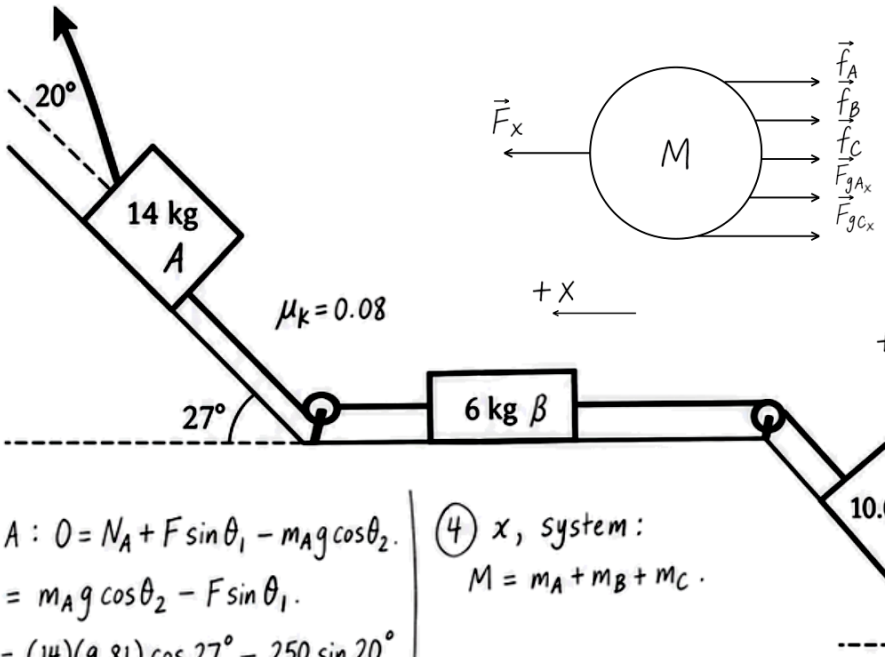
$T_2 = 73.75\ N$

$T_2 = 74\ N$



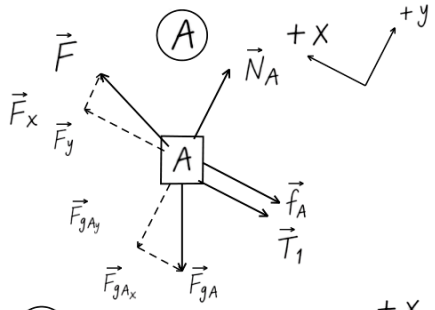
2) A string and pulley system is set up as shown below. The string is massless and inextensible and the pulley frictionless. The coefficient of kinetic friction between the blocks and the surface is 0.08. Calculate:
a) the acceleration of the 6.0 kg block. [2.6 m/s^2]
b) the tension in the string between the 14.0 kg and 6.0 kg blocks. [133 N]

F_{app} = 250 N

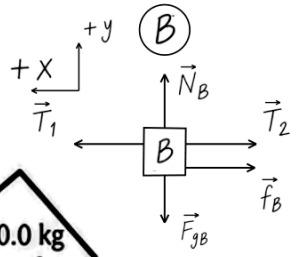


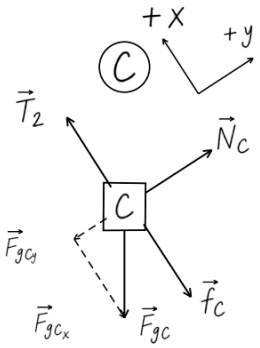
unfolded system

+x: A uphill; B left; C uphill.



+x





① y, A: $0 = N_A + F\sin\theta_1 - m_A g\cos\theta_2$.

$N_A = m_A g\cos\theta_2 - F\sin\theta_1$.

$N_A = (14)(9.81)\cos 27° - 250\sin 20°$

$N_A = 36.87$ N

② y, B: $0 = N_B - m_B g$

$N_B = m_B g$

$N_B = (6)(9.81) = 58.86$ N

③ y, C: $0 = N_C - m_C g\cos\theta_3$

$N_C = m_C g\cos\theta_3$

$N_C = (10.0)(9.81)\cos 57°$

$N_C = 53.43$ N

④ x, system:

$M = m_A + m_B + m_C$.

$F_{net,x} = F\cos\theta_1 - \mu_k(N_A + N_B + N_C) - m_A g\sin\theta_2 - m_C g\sin\theta_3 = Ma$.

$$a = \frac{\left[F\cos\theta_1 - \mu_k\left[m_A g\cos\theta_2 - F\sin\theta_1 + m_B g + m_C g\cos\theta_3\right] - g\left(m_A\sin\theta_2 + m_C\sin\theta_3\right)\right]}{M}.$$

$$a = \frac{250\cos 20° - 0.08(36.87 + 58.86 + 53.43) - 9.81\left(14\sin 27° + 10.0\sin 57°\right)}{14 + 6 + 10.0}$$

$a = +2.61220$ m/s^2

$a = +3$ m/s^2 (1 s.f.)

$\vec{a}_B = 3$ m/s^2 [positive x] (1 s.f.)

+x reversed: A downhill, B right, C downhill.

$$a_2 = \frac{-F\cos\theta_1 - \mu_k(N_A + N_B + N_C) + g\left(m_A\sin\theta_2 + m_C\sin\theta_3\right)}{M} = -3.41 \text{ m/s}^2$$

Opposite to this case's +x.

(b) x, A: $m_A a = F\cos\theta_1 - \mu_k N_A - m_A g\sin\theta_2 - T_1$.

$T_1 = F\cos\theta_1 - \mu_k(m_A g\cos\theta_2 - F\sin\theta_1) - m_A g\sin\theta_2 - m_A a$.

$T_1 = 250\cos 20° - 0.08(36.87) - (14)(9.81)\sin 27° - (14)(2.61220)$

$T_1 = 133.05$ N

$T_1 = 100$ N (1 s.f.)

Homework

Finish the classwork problems (if not done).